Issues Mobile IPv4

By

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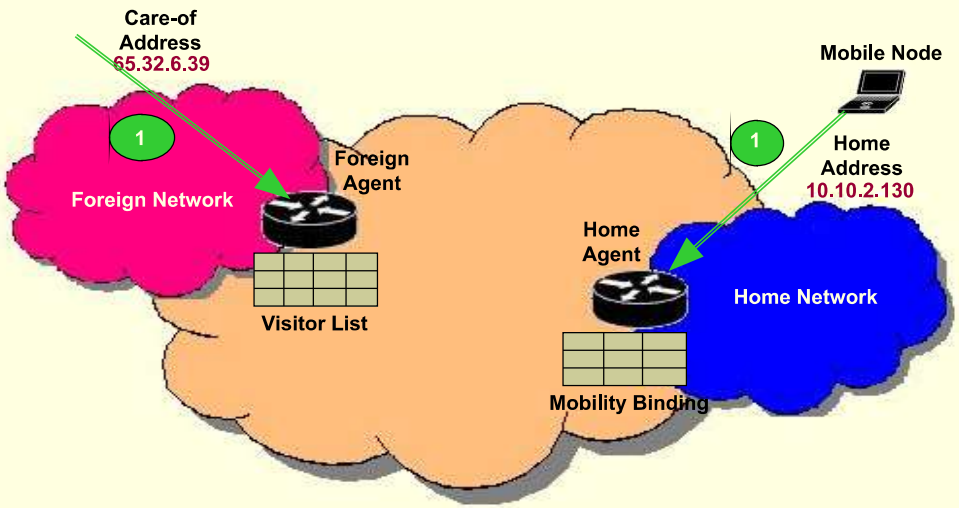
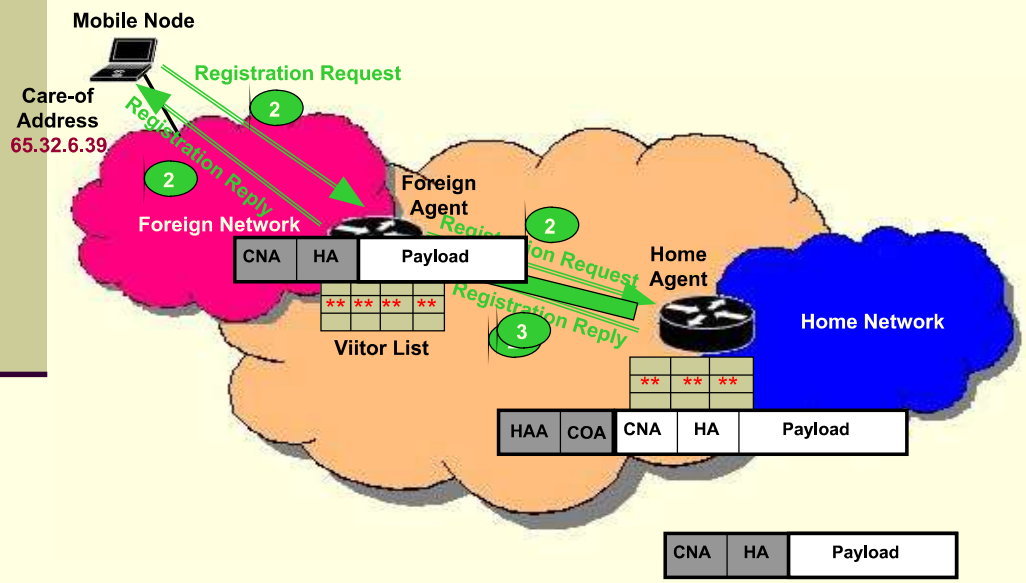
Mobile IPv4



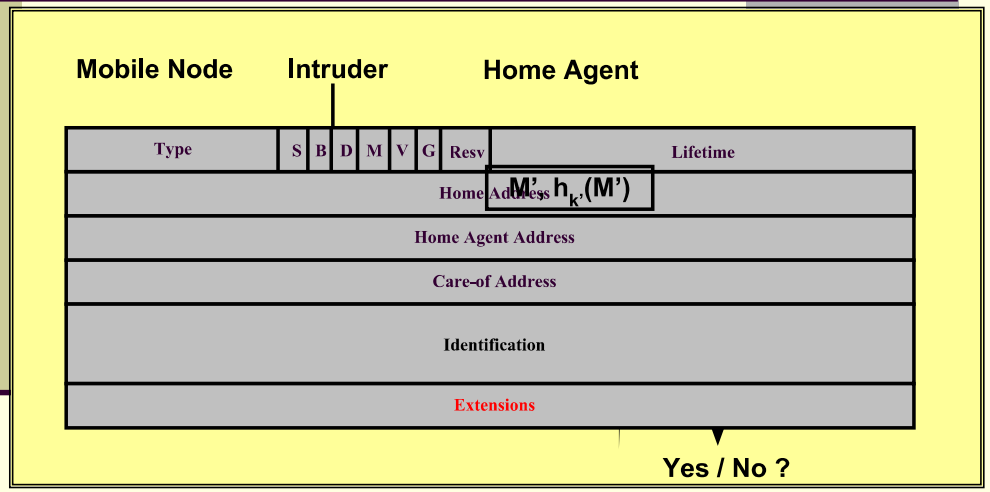
Mobile IPv4



Types of Attacks in Mobile IP

- Denial-of-Service Attack:
  - *An Intruder modify/generates a wrong Registration Request Message specifying his own IP Address to be the Care-of Address of a Mobile Node.*
  - *Thus all the traffic for the Mobile Node will be forwarded to the Intruder.*
- Replay Attack:
  - *An Intruder may obtain a copy of a valid Registration Request from a Mobile Node, store it and then "Replay" it at latter time frequently.*
  - *Thus effectively cutting the Mobile Node from the Network.*

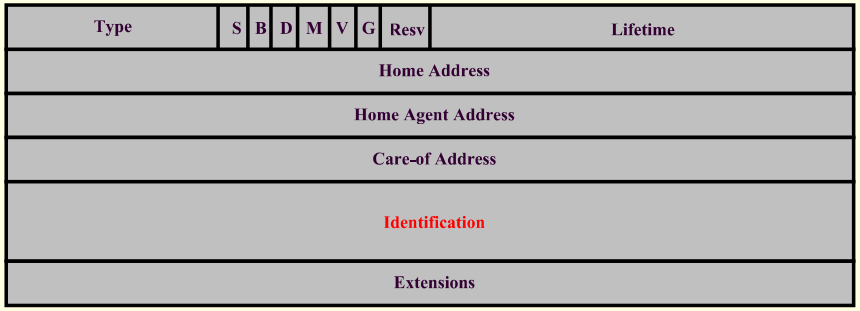
Solution : Denial-of-Service Attack



- It allows the Home Agent to deny the service if the Registration Request Message is modified.

Solution : Replay Attack

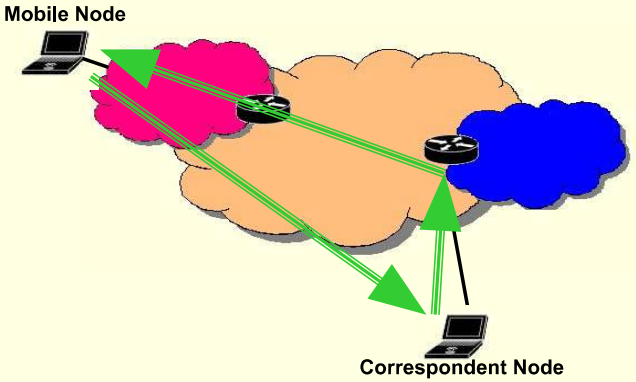
- The Identification field in Registration Request message are implemented with a Pseudo Random Sequences.

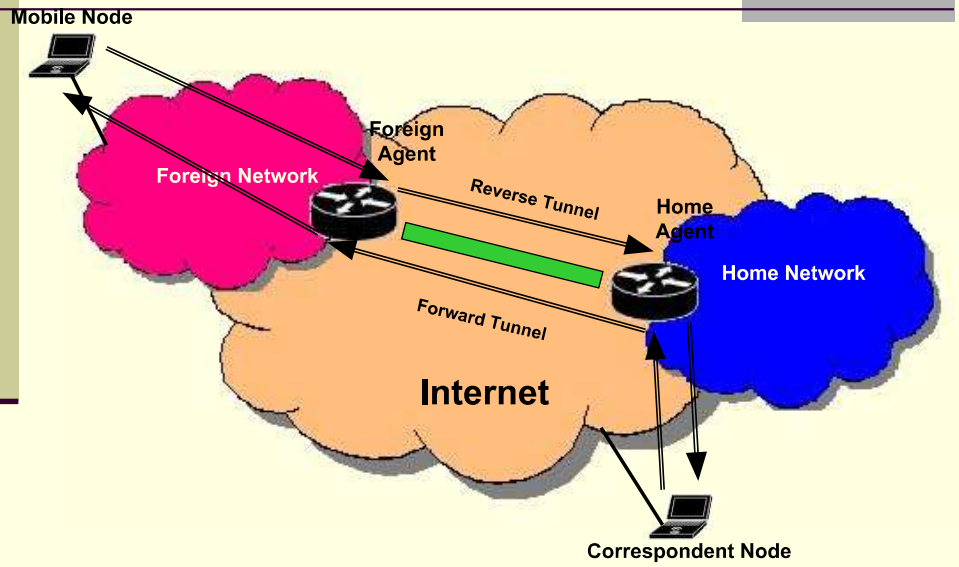


- The next sequence is unknown to the Intruder effectively the Home Agent will through the request from the Intruder.

Triangular Routing

- Traffic from CN to MN is forwarded through HA & traffic from MN is forwarded directly to CN. This kind of routing is referred to as Triangular Routing.



- Problem : Outgoing packet can't use Home Address as the source address. Some anti-spoofing software at the foreign network prevent this.

Reverse Tunneling (Cont.)